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## Calculation

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The system of differential equation for the LC-oscillator is:

$$\begin{aligned} C \frac{dV}{dt} &= I(t), & V(0) &= 0, \\ L \frac{dI}{dt} &= V_B - R I(t) - V(t), & I(0) &= 0. \end{aligned}$$

Constants  $V_B$ ,  $L$  and  $R$  are given:  $V_B = 5$  V,  $L = 0.004$   $\mu$ H,  $R = 0.1$   $\Omega$ , and  $C$  has to be found such that the circuit oscillates with at frequency of 100 MHz =  $10^8$  Hz.

First, we need to choose a time scale, and then we have to scale the problem and choose an appropriate step size.

### Exercise A

1/1 point (graded)

First, a clock frequency of 100 MHz, means that one oscillation has a period of  $10^{-8}$  seconds.

What is a handy time scale for this problem?

☐ seconds

☒ nanoseconds ✓

☐ gigaseconds

Submit

You have used 1 of 1 attempt

**i** Answers are displayed within the problem

The differential equations above, have been defined for time  $t$  in seconds. For the numerical calculations that is not very convenient, as you would have to calculate with very small numbers: the tiny period, and the even smaller inductance. Therefore, we will first scale the time  $t$  to  $\tau$ , such that when  $t$  is in seconds,  $\tau$  is in nanoseconds:  $\tau = 10^9 t$ .

In the differential equations,  $t$  only has to be changed into  $\tau$  in the two derivatives. For that you use the chain rule for differentiation:  $\frac{df}{dt} = \frac{df}{d\tau} \frac{d\tau}{dt} = \frac{df}{d\tau} 10^9$ . So the differential equations become:  $10^9 C \frac{dV}{d\tau} = I$  and  $10^9 L \frac{dI}{d\tau} = V_B - RI - V$ . Then the factor  $10^9$  can conveniently be included in  $L$  and  $C$ , so these become manageable as well:

$$\begin{aligned}\hat{C} \frac{dV}{d\tau} &= I, & V(0) &= 0 \\ \hat{L} \frac{dI}{d\tau} &= V_B - RI - V, & I(0) &= 0,\end{aligned}$$

where  $\hat{C} = 10^9 C$  and  $\hat{L} = 10^9 L = 4$ . And still  $V_B = 5$  V and  $R = 0.1 \Omega$ .

As a first estimation of  $C$  we will use  $C = 0.5 \times 10^{-9}$  farad, so  $\hat{C} = 0.5$ .

## Exercise B

1/1 point (graded)

Make a fresh copy of your basic Euler's method simulation program for systems and implement the initial value problem for the circuit. Take  $\Delta\tau = 1$  nanosecond and simulate 50 nanoseconds of the solutions ( $t_{end} = 50$ ). We will only look at the solutions for the current.

When  $\Delta\tau = 1$ ,  $I(10) \approx$

Give your answer in 4 decimals.

✓ Answer: -0.8598



You have used 1 of 3 attempts

❗ Answers are displayed within the problem

## Exercise C

1/1 point (graded)

The current  $I(t)$  is obviously unrealistic when simulated with  $\Delta\tau = 1$ . Try  $\Delta\tau = 0.1$ ,  $\Delta\tau = 0.01$ ,  $\Delta\tau = 0.001$ , etc..

What is the largest of the above values for  $\Delta\tau$ , for which the results are not obviously wrong?

✓ Answer: 0.01

You have used 2 of 3 attempts

📘 Answers are displayed within the problem

## Exercise D

2/2 points (graded)

Use  $\Delta\tau = 0.01$  and estimate the current after 2.2 nanoseconds:  $I(2.2)$ . Simulate again for  $\Delta\tau = 0.02$  and with this new estimate for  $I(2.2)$ , estimate the error. Give all your answers in four decimals. (Do not forget that an error can have a sign!)

Hint: you can check whether you have the right index, by looking up the time for that index in the time array.

For  $\Delta\tau = 0.01$ ,  $I(2.2) \approx$

✓ Answer: 1.7294

For  $\Delta\tau = 0.02$ ,  $I(2.2) \approx$

✓ Answer: 1.7389

The error in  $I(2.2)$  for  $\Delta\tau = 0.01$ , can be estimated as

✓ Answer: -0.00954

You have used 1 of 3 attempts

**i** Answers are displayed within the problem

## Exercise E

1/1 point (graded)

Take  $\Delta\tau = 0.001$  and calculate  $I(2.2)$  in at least 5 decimals, and estimate the error by recalculating for  $\Delta\tau = 0.002$  in 5 decimals. Show that the error is smaller than **0.001**.

The error in  $I(2.2)$ , when  $\Delta\tau = 0.001$ , can be estimated as  
Use 5 decimals.

✓ Answer: -0.00095

You have used 2 of 3 attempts

**i** Answers are displayed within the problem

## Exercise F

1/1 point (graded)

With  $\Delta\tau = 0.001$  and  $t_{\text{end}} = 50$  ns, estimate the period of the solution (the length of the cycles).

The period of the oscillation in  $I(t) \approx$   
Give your answer in nanoseconds with one decimal.

✓ Answer: 8.9

You have used 1 of 3 attempts

**i** Answers are displayed within the problem

## Exercise G

1/1 point (graded)

We finally have the tools to answer the question: What value of  $\hat{C}$  is best to obtain oscillations with a frequency of **100** MHz, so with periods of **10** nanoseconds? Choose values for  $\hat{C}$  near **0.5** until you tune in on the best value in two decimals.

What is the value of  $\hat{C}$ ?

Give your answer in 2 decimals.

✓ Answer: 0.633

You have used 1 of 3 attempts

**i** Answers are displayed within the problem

On the next page, you can reflect on the results you've got now, and investigate what happens when you have different values of  $R$ .

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 English ▼

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